Kamruzzaman Joty

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Summary

PhD candidate in Mechanical Engineering (Southern Methodist University, expected May 2027) with 3+ years of hands-on research in nanofabrication and single-molecule electrical measurement. Formed nanopores from 2.6 to about 25 nm in 12 nm SiN_x membranes by controlled dielectric breakdown, including through stacked 2D materials (graphene, MoS_2, hBN), and analyzed datasets of more than 565,000 translocation events with Python. Co-author of 8 peer-reviewed papers (4 as first or co-first author) and 1 U.S. provisional patent application. Also designed 3D-printed hardware for a low-cost nanopore fabrication platform.

Research Experience

Graduate Research Assistant, BAST Laboratory Jan 2023 – Present

Department of Mechanical Engineering, Southern Methodist University, Dallas, TX

Multilayer 2D-material nanopores (SiN_x / graphene / MoS_2 / hBN)

- Dry-transferred one or two 2D layers (hBN, MoS_2, graphene) onto 12 nm SiN_x chips, then formed ~20 nm pores by chemically tuned controlled dielectric breakdown (CT-CDB, a method developed earlier in our lab): pore formation in 10 minutes or less per device, about 78% device yield across 46 chips (project total). Verified layer presence at the pore region by SEM-EDX.
- Tested devices with holo-transferrin translocation. Across more than 565,000 events (first author, *ACS Applied Nano Materials*), the terminal 2D layer set the polarity of directional slowing in asymmetric pores: hBN and MoS_2 gave opposite directions.
- Applied Gaussian mixture modeling and effect-size statistics in Python to show direction-dependent shifts in event subpopulations (IEEE DCAS 2026, first author).
- Developed a stacked-ensemble machine-learning classifier that identified membrane architecture at 96% accuracy across six architectures (>300,000 events; *Small* 2026).

P.O.R.E. (Programmable Ohmic Research Engine)

- Mechanical and fluidic design of a USB-powered, smartphone-sized platform (110 × 75 × 40 mm) for in situ nanopore fabrication, with core electronics under $100: CAD modeling (Autodesk Inventor), FDM 3D printing, and assembly of the enclosure and flow-cell cartridge (*IEEE Access* 2026).
- Platform validated at 70–95% pore-formation yield for 6–20 nm targets (80 attempts); redesigned a plug-and-play flow-cell cartridge for the setup (Nature Protocols manuscript in preparation).

Hybrid nanopores with biological components

- Developed α-hemolysin-integrated nanopipettes (sub-10 nm tip) with reproducible protein capture, reversible release, and single-molecule ssDNA transport; designed a custom flow cell for nanopipette measurements (manuscript in preparation).
- Trapped octahedral DNA origami in ~25 nm solid-state pores; for holo-transferrin, dwell time roughly doubled (197 vs 95 µs) and event area more than doubled (9.4 vs 4.4 fC) compared with open pores (*Analytical Chemistry* 2024).
- Fabricated 2.6–5.6 nm pores and measured DNA-templated silver nanoclusters at 200–400 mV, showing the clusters stay intact in KNO_3 and dissociate in KCl (*ACS Applied Materials & Interfaces* 2025).

Research Techniques

Fabrication and processing: dry transfer of 2D materials onto SiN_x membranes; CT-CDB pore formation (2.6 to ~25 nm) in 12 nm SiN_x; sub-10 nm nanopipette fabrication (Sutter P-2000 laser puller); BOE etching; piranha and plasma cleaning of SiN_x chips; gold sputtering; electrolyte and biomolecule preparation.

Measurement and analysis: single-molecule ionic-current recording (Axopatch 200B, Digidata, pCLAMP; 250 kHz sampling); event extraction (EventPro); Python (feature extraction, Gaussian mixture models, effect-size statistics, machine learning); OriginPro; MATLAB.

Instrument design: CAD (Autodesk Inventor), FDM 3D printing, and assembly of enclosures and flow cells.

Familiarity with instruments: JEOL SEM with EDX (nanopipette imaging, layer verification); Park Systems NX-12 AFM (lipid nanoparticle characterization).

Exposure through coursework (ECE graduate cleanroom labs): photolithography, photoresist processing, mask alignment, thermal oxidation, etching, metallization.

Other tools: COMSOL Multiphysics; ANSYS Fluent; AutoCAD; LabVIEW.

Education

Ph.D., Mechanical Engineering Expected May 2027
Southern Methodist University, Dallas, TX | GPA 3.92/4.00
Dissertation: *Hybrid Nanopores for Next-Generation Single-Molecule Sensing*
Relevant graduate coursework: Semiconductor Devices and Fabrication; Compound Semiconductor Devices and Processes; Integrated Photonics; Manufacturing Methods and Systems; Machine Learning Using Python.

B.Eng., Mechanical Engineering Feb 2022
SEGi University, Selangor, Malaysia | GPA 3.87/4.00
Undergraduate Thesis: *Numerical Investigation of Onera M6 Forward and Backward Swept C-Wing Configuration at Supersonic Speed*

Publications and Patent

8 peer-reviewed journal papers (4 as first or co-first author), 2 manuscripts under review, 3 conference proceedings, and 1 U.S. provisional patent application. Full list available on request. First and co-first author papers:

- **Joty, K.***, Gu, C.*, et al. Heterogeneous Multilayer Nanopores via Chemically Tuned Dielectric Breakdown for Single-Molecule Sensing. *Small* 2026, 22(36), e13242. (*equal contribution)
- **Joty, K.**, Gu, C., Lee, S., Kim, M. J. Directional Single-Protein Transport Enabled by 2D Material Heterointerfaces in Solid-State Nanopores. *ACS Applied Nano Materials* 2026, 9(13), 5988–5999.
- **Joty, K.**, Hong, S., et al. Solid-State Nanopore Analysis of DNA-Hosted Silver Nanoclusters: Voltage-Dependent Translocation and Electrolyte Stability. *ACS Applied Materials & Interfaces* 2025, 17(27), 39407–39419. (Supplementary cover art)
- **Joty, K.**, Ghimire, M. L., et al. DNA Origami Incorporated into Solid-State Nanopores Enables Enhanced Sensitivity for Precise Analysis of Protein Translocations. *Analytical Chemistry* 2024, 96(44), 17496–17505. (Supplementary cover art)

First-author conference paper: Joty, K., Gu, C., Lee, S., Kim, M. J. Direction-Aware Signal Analysis for Protein Translocation in Multilayer Solid-State Nanopores. IEEE DCAS 2026.
Also co-author: IEEE Access (P.O.R.E.), Nano Letters, Advanced Materials Interfaces, Electrophoresis.
Patent application: Kim, M. J.; Joty, K.; Gu, C. “Heterogeneous Multilayer Nanopores.” U.S. Provisional Application No. 63/998,374, filed September 28, 2026 (pending).

Teaching and Mentoring

- **Graduate Teaching Assistant**, Department of Mechanical Engineering, SMU (2 semesters, 2024–2025): lab instructor for classes of about 20 students in Fluid Mechanics, Thermodynamics, Heat Transfer, and Nano-Biomanufacturing Laboratory.
- Mentored 2 undergraduate researchers in the BAST Laboratory and 10 high-school students (Governor’s Science and Technology Champions’ Academy, 2025) in nanopore experiments and data analysis.

Honors and Service

Honors: 2nd place, Outstanding Research & Presentation Award, SMU Industry Innovation Summit (2026); People’s Choice Award, SMU Three Minute Thesis (2025); Best Presentation Award, SMU Research & Innovation Week (2024); Lyle Tuition Fellowship, SMU (2023–present); IMechE Project Award (2022).
Service: Lyle Senator, SMU Student Senate (2025–2026); peer reviewer (2 reviews), *ACS Omega* (2026).

Additional Experience

Advising Officer, SA Classic Properties Sdn. Bhd., Kuala Lumpur, Malaysia Mar – Dec 2022
Industrial Trainee, Bangladesh-India Friendship Power Company Ltd., Bangladesh Jun – Sep 2021